

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
[varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_Lab
varnikakomali@VARNIKAs-MacBook-Air OS_Lab % cd Practical
varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical5.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical5.c -o practical5
varnikakomali@VARNIKAs-MacBook-Air Practical % ./practical5
Child (Consumer) received 10240000 bytes.
Parent (Producer) finished sending data.
Total Data Transferred: 9.77 MB
Time Elapsed: 0.008802 seconds
Communication Efficiency (Throughput): 1109.48 MB/s
varnikakomali@VARNIKAs-MacBook-Air Practical %
```

```
[varnikakomali@VARNIKAs-MacBook-Air Practical % nano pipe_cmd.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc pipe_cmd.c -o pipe_cmd && ./pipe_cmd
-rw-r--r-- 1 varnikakomali staff 54 3 Aug 19:45 copy_sample.txt
-rw-r--r-- 1 varnikakomali staff 295 17 Aug 10:35 permission.c
-rwxr-xr-x 1 varnikakomali staff 33848 31 Aug 08:35 pipe_cmd
-rw-r--r-- 1 varnikakomali staff 986 31 Aug 08:35 pipe_cmd.c
-rwxr-xr-x 1 varnikakomali staff 33912 14 Aug 14:27 practical4
-rw-r--r-- 1 varnikakomali staff 1642 14 Aug 14:26 practical4.c
-rwxr-xr-x 1 varnikakomali staff 33944 31 Aug 08:33 practical5
-rw-r--r-- 1 varnikakomali staff 1603 31 Aug 08:33 practical5.c
-rw-r--r--@ 1 varnikakomali staff 910 3 Aug 11:56 question1.c
-rw-r--r--@ 1 varnikakomali staff 910 3 Aug 11:16 question1.c
-rw-r--r-- 1 varnikakomali staff 785 3 Aug 20:06 question2.c
-rw-r--r-- 1 varnikakomali staff 1230 8 Aug 14:51 question3.c
varnikakomali@VARNIKAs-MacBook-Air Practical %
```